

Docker Task -3

Task Description:

Create a custom docker image for nginx and deploy it using docker compose, where the volume bind mount should be at /var/opt/nginx location. Push the created custom docker image to your docker-hub.

Docker and Docker compose installation :

```
ubuntu@ip-172-31-39-36:~$ sudo apt update -y
sudo apt install -y docker.io
sudo systemctl start docker
sudo systemctl enable docker
sudo usermod -aG docker ubuntu
lnnewgrp docker
Hit:1 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute InRelease
Hit:2 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:3 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-backports InRelease
Hit:4 https://download.docker.com/linux/ubuntu resolute InRelease
Hit:5 http://security.ubuntu.com/ubuntu resolute-security InRelease
61 packages can be upgraded. Run 'apt list --upgradable' to see them.
The following package was automatically installed and is no longer required:
  docker-ce-rootless-extras
Use 'sudo apt autoremove' to remove it.

Installing:
  docker.io

Installing dependencies:
  bridge-utils  containerd  dns-root-data  dnsmasq-base  runc  ubuntu-fan
```

```
ubuntu@ip-172-31-39-36:~$ sudo curl -L "https://github.com/docker/compose/releases/latest/download/docker-compose-$(uname -s)-$(uname -m)" -o /usr/local/bin/docker-compose
sudo chmod +x /usr/local/bin/docker-compose
docker-compose --version
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total   Spent    Left   Speed
  0     0    0     0  0     0     0     0     0     0
  0     0    0     0  0     0     0     0     0     0
100 31.05M 100 31.05M    0     0  46.84M    0     0     0
Docker Compose version v5.3.1
```

Created index.html file :

```
timed11 -p nginx content
ubuntu@ip-172-31-39-36:~/nginx-docker$ echo "<h1>Hello from Custom Nginx on AWS EC2!</h1>" > nginx-content/index.html
ubuntu@ip-172-31-39-36:~/nginx-docker$
```

Dockerfile :

```
restart: always
[ubuntu@ip-172-31-39-36:~/nginx-docker$ cat Dockerfile
FROM nginx:latest

LABEL maintainer="rajgovindarajan48"

# Copy custom HTML content into nginx's default serve location
COPY nginx-content/ /usr/share/nginx/html/

# Expose port 80
EXPOSE 80

CMD ["nginx", "-g", "daemon off;"]
```

docker-compose.yml file :

```
[ubuntu@ip-172-31-39-36:~/nginx-docker$ cat docker-compose.yml
version: '3.8'

services:
  nginx:
    image: rajgovindarajan48/custom-nginx:latest
    build:
      context: .
      dockerfile: Dockerfile
    container_name: custom-nginx-container
    ports:
      - "80:80"
    volumes:
      - ./nginx-content:/var/opt/nginx # bind mount at /var/opt/nginx
    restart: always
```

Docker login :

```
[ubuntu@ip-172-31-39-36:~/nginx-docker$ docker login

USING WEB-BASED LOGIN

Info → To sign in with credentials on the command line, use 'docker login -u <username>'

Your one-time device confirmation code is: HNNC-HNXF
Press ENTER to open your browser or submit your device code here: https://login.docker.com/activate

Waiting for authentication in the browser...

WARNING! Your credentials are stored unencrypted in '/home/ubuntu/.docker/config.json'.
Configure a credential helper to remove this warning. See
https://docs.docker.com/go/credential-store/

Login Succeeded
```

Docker Build :

```
[ubuntu@ip-172-31-39-36:~/nginx-docker$ docker build -t rajgovindarajan48/custom-nginx:latest .
[+] Building 6.1s (8/8) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring Dockerfile: 272B
=> [internal] load metadata for docker.io/library/nginx:latest
=> [auth] library/nginx:pull token for registry-1.docker.io
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [internal] load build context
=> => transferring context: 133B
=> [1/2] FROM docker.io/library/nginx:latest@sha256:b5a9a3cfc8b681dd6f1458d53f2ad2b559cc255653d2efdcacee5636bf2066f
=> => resolve docker.io/library/nginx:latest@sha256:b5a9a3cfc8b681dd6f1458d53f2ad2b559cc255653d2efdcacee5636bf2066f
=> sha256:45ef33eb91c8a1ddf8e57a108f3444dc5515bcb8e81bb7b69ba92eab66c1 1.40kB / 1.40kB
=> sha256:6a24a6ebcf38e20b7399e8e49d114d1cc641afecb94382c93a53239b2cd680dd 1.21kB / 1.21kB
=> sha256:8d2892eabbc5d47a861f8a6c9d8481abeb8b2975095a583bfe9ea9f6593cab 403B / 403B
=> sha256:0b19be7179947081f6a48f13c58423b1427852de2546ba583f036df572ce10d 956B / 956B
=> sha256:3b9f359a82680768cbbae872a2ecb68eb887285247d2fab6e9e89e8d9121ff9 628B / 628B
=> sha256:3b9f359a82680768cbbae872a2ecb68eb887285247d2fab6e9e89e8d9121ff9 628B / 628B
=> sha256:062e458697faa5f02a3a74eba986aee4d79bc9cfd65769fc6dcdf2c05c6a853 29.78MB / 29.78MB
=> extracting sha256:062e458697faa5f02a3a74eba986aee4d79bc9cfd65769fc6dcdf2c05c6a853 1.05s
=> extracting sha256:3b9f359a82680768cbbae872a2ecb68eb887285247d2fab6e9e89e8d9121ff9 1.05s
=> extracting sha256:0b19be7179947081f6a48f13c58423b1427852de2546ba583f036df572ce10d 0.05s
=> extracting sha256:4a24a6ebcf38e20b7399e8e49d114d1cc641afecb94382c93a53239b2cd680dd 0.05s
=> extracting sha256:45ef33eb91c8a1ddf8e57a108f3444dc5515bcb8e81bb7b69ba92eab66c1 0.05s
=> [2/2] COPY nginx-content/ /usr/share/nginx/html/
=> exporting to image
=> exporting layers
=> exporting manifest sha256:440940dc949f71238536867422c9dee7a5e5c51c1afba1582cfe0c33b2155b 0.15s
=> exporting config sha256:b22813e5acfdad1011ea01b40749b598db93f5b13685788bd0f7a7ebca5895a53 0.05s
=> exporting attestation manifest sha256:20b26726bcd0c96be39ff9070e6251f3016b72e1ee87243eba6d2fcb4a855ba2 0.05s
=> exporting manifest list sha256:c7537dfb403767620b33f034405e73087c68632bb93ab71e479bc094eb6cee84 0.05s
=> naming to docker.io/rajgovindarajan48/custom-nginx:latest 0.05s
=> unpacking to docker.io/rajgovindarajan48/custom-nginx:latest 0.05s
```

Docker compose up :

```
[ubuntu@ip-172-31-39-36:~/nginx-docker$ docker-compose up
WARN[0000] /home/ubuntu/nginx-docker/docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] up 1/1
  Image your-dockerhub-username/custom-nginx:latest pull access denied for your-dockerhub-username/custom-nginx, repository does not exist or may require 'docker login' 1.1s
[+] Building 1.2s (9/9) FINISHED
=> [internal] load local bake definitions
=> => reading from stdin 53B
=> [internal] load build definition from Dockerfile
=> => transferring Dockerfile: 272B
=> [internal] load metadata for docker.io/library/nginx:latest
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [internal] load build context
=> => transferring context: 78B
=> [1/2] FROM docker.io/library/nginx:latest@sha256:b5a9a3cfc8b681dd6f1458d53f2ad2b559cc255653d2efdcacee5636bf2066f
=> => resolve docker.io/library/nginx:latest@sha256:b5a9a3cfc8b681dd6f1458d53f2ad2b559cc255653d2efdcacee5636bf2066f
=> CACHED [2/2] COPY nginx-content/ /usr/share/nginx/html/
=> exporting to image
=> exporting layers
=> exporting manifest sha256:c5ab682049a985702bb2c8d6de724389f5c33e06186bf5d5cc69eaf8cabcb19 0.05s
=> exporting config sha256:7d27b6e0c50808ba9573b8d5c5462cc214042c630fca534536b482f6fb7cf29 0.05s
=> exporting attestation manifest sha256:4867beb735786db5081cc344d3660be27215ff9d4bed56b1600e392ebff8e02 0.05s
=> exporting manifest list sha256:7995834e0ed6700776c28a5771d79b65b10efee1834b26a6b221377e0842b058 0.05s
=> naming to docker.io/your-dockerhub-username/custom-nginx:latest 0.05s
[+] up 3/3acking to docker.io/your-dockerhub-username/custom-nginx:latest
  Image your-dockerhub-username/custom-nginx:latest Built 2.3s
  Network nginx-docker_default Created 0.1s
  Container custom-nginx-container Started 0.05s
```

Docker push

```
[ubuntu@ip-172-31-39-36:~/nginx-docker]$ docker push rajgovindarajan48/custom-nginx:latest
The push refers to repository [docker.io/rajgovindarajan48/custom-nginx]
5d7738a58034: Pushed
e99022f7d018: Pushed
3b9f3509a826: Pushed
0b19be717994: Pushed
39694cde734e: Pushed
8d2092eabbc5: Pushed
062e450697fa: Pushed
45ef33eb91c8: Pushed
6a24a6ebcf30: Pushed
latest: digest: sha256:c7537dfb403767620b33f034605e730876d8632bb93ab71e479bc094eb6cee04 size: 856
```

Docker compose :

```
[ubuntu@ip-172-31-39-36:~/nginx-docker]$ docker-compose up -d
WARN[0000] /home/ubuntu/nginx-docker/docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] up 1/1
  Image your-dockerhub-username/custom-nginx:latest pull access denied for your-dockerhub-username/custom-nginx, repository does not exist or may require 'docker login'      1.1s
[+] Building 1.2s (9/9) FINISHED
=> [internal] load local bake definitions                                0.0s
=> => reading from stdin 5338                                           0.0s
=> [internal] load build definition from Dockerfile                     0.0s
=> => transferring dockerfile: 272B                                     0.0s
=> [internal] load metadata for docker.io/library/nginx:latest         0.7s
=> [internal] load .dockerignore                                         0.0s
=> => transferring context: 2B                                           0.0s
=> [internal] load build context                                         0.0s
=> => transferring context: 78B                                          0.0s
=> [1/2] FROM docker.io/library/nginx:latest@sha256:b5a9a3cfc86b81dd6f1458d53f2ad2b559cc255653d2efdcacae5636bf2066f  0.0s
=> => resolve docker.io/library/nginx:latest@sha256:b5a9a3cfc86b81dd6f1458d53f2ad2b559cc255653d2efdcacae5636bf2066f  0.0s
=> CACHED [2/2] COPY nginx-content/ /usr/share/nginx/html/            0.0s
=> => exporting image                                                    0.1s
=> => exporting layers                                                    0.0s
=> => exporting manifest sha256:c5ab6820a9a985782b2c86ddee724389f5c33e06186bf5d5cc69aefca4bc19  0.0s
=> => exporting config sha256:7927b6dede0808a9973b8d5c5442c214042c30fca53453db482f6f4b7cf29  0.0s
=> => exporting attestation manifest sha256:4867bab73578db5801cc344d36a0be27215ff94bed56be1a00a392ebff8e2  0.0s
=> => exporting manifest list sha256:7995834e0ed6780776c28a5771d79b65b10efee1834b266ab221377e8042b058  0.0s
=> => naming to docker.io/your-dockerhub-username/custom-nginx:latest  0.0s
[+] up 3/3 Backing to docker.io/your-dockerhub-username/custom-nginx:latest  0.0s
  Image your-dockerhub-username/custom-nginx:latest built              2.3s
  Network nginx-docker_default Created                                 0.1s
  Container custom-nginx-container Started                             0.5s
```

Docker log and index.html :

```
[ubuntu@ip-172-31-39-36:~/nginx-docker]$ docker-compose logs nginx
WARN[0000] /home/ubuntu/nginx-docker/docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion
custom-nginx-container | /docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
custom-nginx-container | /docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
custom-nginx-container | /docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
custom-nginx-container | 10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
custom-nginx-container | 10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
custom-nginx-container | /docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
custom-nginx-container | /docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
custom-nginx-container | /docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
custom-nginx-container | /docker-entrypoint.sh: Configuration complete; ready for start up
custom-nginx-container | 2026/07/15 11:32:15 [notice] 1#1: using the "epoll" event method
custom-nginx-container | 2026/07/15 11:32:15 [notice] 1#1: nginx/1.31.2
custom-nginx-container | 2026/07/15 11:32:15 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
custom-nginx-container | 2026/07/15 11:32:15 [notice] 1#1: OS: Linux 7.0.0-1006-aws
custom-nginx-container | 2026/07/15 11:32:15 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1024:524288
custom-nginx-container | 2026/07/15 11:32:15 [notice] 1#1: start worker processes
custom-nginx-container | 2026/07/15 11:32:15 [notice] 1#1: start worker process 30
custom-nginx-container | 2026/07/15 11:32:15 [notice] 1#1: start worker process 31
[ubuntu@ip-172-31-39-36:~/nginx-docker]$ docker exec -it custom-nginx-container ls /var/opt/nginx
index.html
```

nginx server is accessed using public IP :



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rajgovindarajan48/custom-nginx

Last pushed about 1 hour ago • Repository size: 60.2 MB • ☆ 0 • ↓ 19

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